

Thursday Reading Assessment: ADC and DAC (Chapter 12 + Lab)

Prof. Jordan C. Hanson

April 11, 2024

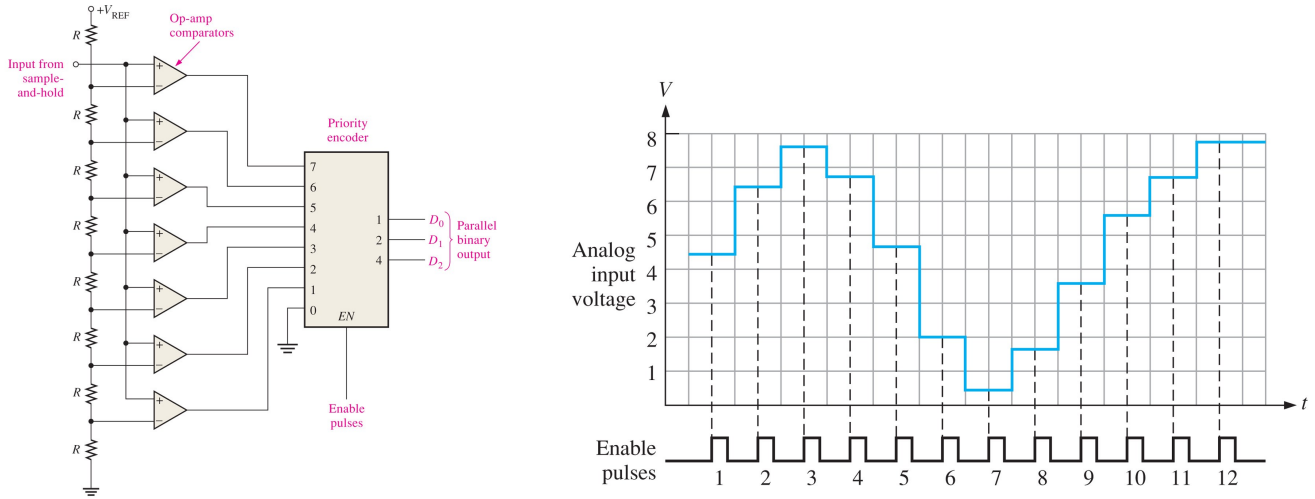


Figure 1: (Left) The 3-bit **Flash ADC**. (Right) The input to the 3-bit Flash ADC.

1 Analog-to-Digital Conversion

- For the ADC in Fig. 1 (left), $V_{ref} = +8$ Volts. (a) What is the voltage *resolution*? (b) Determine the binary code output of the 3-bit flash ADC in Fig. 1 (left), given the input signal in Fig. 1 (right), given the encoder enable pulses shown. (c) If the enable pulse frequency in Fig. 1 (right) were halved, determine the binary numbers represented by the resulting digital output. Is any information lost?